

4. (a) Crude oil is a fossil fuel and is described as a non-renewable resource.

(i) Describe how crude oil was formed.

[2]

.....

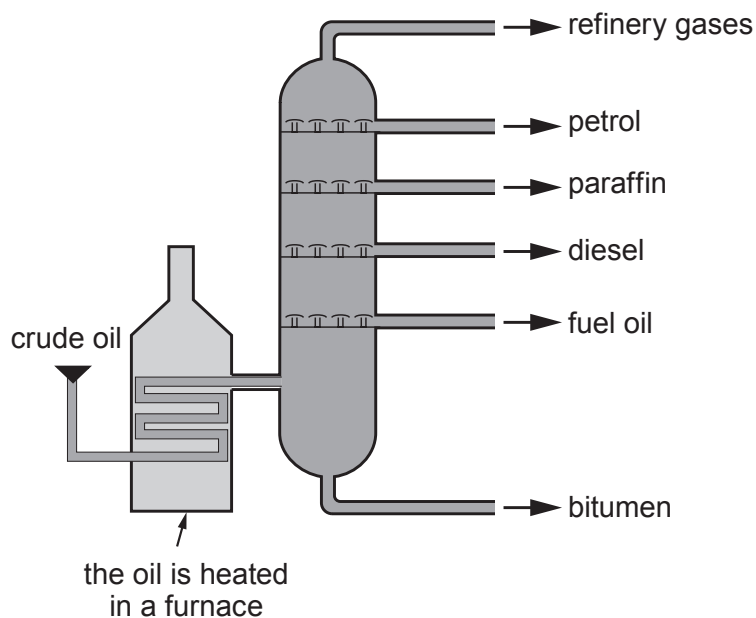
.....

(ii) Give the meaning of the term *non-renewable*.

[1]

.....

(b) To make crude oil more useful, it is separated into fractions.



(i) Complete the following sentences.

[2]

Crude oil is separated into different fractions by a process called fractional

.....

The fractions can be separated because they have different

.....

(ii) **Circle** the word which best describes crude oil.

[1]

mixture

element

compound



- (c) One of the fractions obtained from crude oil contains hexane, C_6H_{14} .

Calculate the percentage by mass of carbon in hexane.

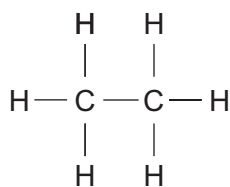
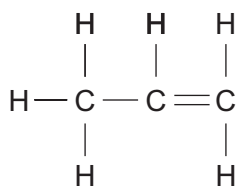
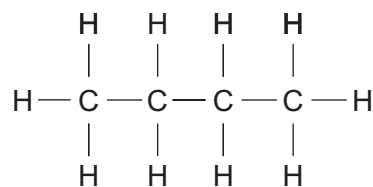
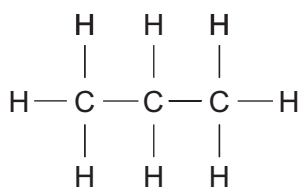
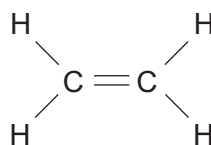
[2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = %



(d) The diagram shows the structures of five different hydrocarbons, **A-E**.

**A****B****C****D****E**

Use letters **A-E** in your answers to parts (i) and (ii).

(i) Give the structure that represents propene. [1]

(ii) Identify the structures that fit the following descriptions. [2]

Hydrocarbons with the general formula $\text{C}_n\text{H}_{2n+2}$

Unsaturated molecules



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- (e) Plastics are made from chemicals that are obtained from crude oil. Supermarkets in Wales were the first in the UK to charge their customers for plastic bags. This was to reduce the amount of plastic waste generated.

Give **two** methods of plastic waste disposal that lead to environmental problems. Explain the problem linked to each method. [2]

Method 1

Problem

.....

Method 2

Problem

.....

13

3430U501
11



4. An alloy is a material composed of a mixture of elements, at least one of which is a metal. The table lists the composition and common uses of different alloys containing silver and gold.

Name of alloy	Composition of alloy by mass (%)	Uses of alloy
amalgam	mercury 48 %, silver 25 %, tin 15 %, copper 12 %	dental fillings, mining
green gold	gold 75 %, silver 6-24 %, copper %	Nobel Prize medals, decoration
nordic gold	gold 89 %, aluminium 5 %, zinc 5 %, tin 1 %	coins, decoration
solder	tin 90 %, silver 5 %, copper 5 %	joining electrical components
sterling silver	silver 92.5 %, platinum 4 %, germanium 1.5-3 %, zinc 0.5-2 %	decoration, plumbing, instruments, jewellery
white gold	gold 75 %, palladium 10 %, nickel 10 %, zinc 5 %	decoration, jewellery

- (a) **Circle** the number of alloys that contain an element from Group 4 of the Periodic Table.

[1]

0 1 2 3 4 5 6

- (b) **Circle** the correct percentage range for the mass of copper that can be found in green gold.

[1]

0-25 % 1-19 % 19-25 % 6-24 %



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(c) Tick (✓) the statement that **best** describes the composition of the alloys listed. [1]

- the alloys all contain at least one metal ☐
- the alloys all contain at least two metals ☐
- the alloys all contain at least three metals ☐
- the alloys all contain at least four metals ☐

(d) Tick (✓) the statement that **best** describes the decorative uses of the alloys listed. [1]

- all of the alloys are used for decorative purposes ☐
- all of the alloys containing gold are used for decorative purposes ☐
- all of the alloys containing silver are used for decorative purposes ☐
- none of the alloys containing silver are used for decorative purposes ☐

(e) A solder joint in an electrical circuit contains 0.00011 kg of silver.
Use this information and the composition of solder given in the table to calculate the mass of tin in the solder joint. [2]

Mass = kg

6



4. (a) Supermarkets across the UK now charge customers a minimum of 5p for every plastic carrier bag supplied.

The reason for the charge is to reduce the number of plastic carrier bags used by consumers and therefore reduce waste and litter.

Since the introduction of the charge for plastic carrier bags in Wales, the number of these bags used by consumers has fallen by over 70 %.

Similarly, reports in England claim that supermarkets have issued 83 % fewer bags since the charge was introduced.

It is estimated that every person in the UK currently uses around 25 plastic carrier bags per year, compared to around 140 before charges were introduced.

The majority of the money retailers generate from sales of carrier bags is donated to good causes. A survey of retailers across England and Wales reported that £87 million had been donated to good causes since the introduction of the 5p charge, amounting to 4p for every bag sold.



- (i) Use the information in the passage to decide whether the following statements are **true** or **false**.

Put a tick (✓) in the correct column for each statement.

[3]

Statement	True	False
The number of plastic bags used in Wales and England has reduced since charging for them		
Retailers donate all the money generated from the sale of plastic bags to good causes		
Plastic bags are no longer used		
The charge for plastic bags has totally stopped their use in Wales		
The use of plastic bags leads to environmental problems		
The charge for plastic bags is beneficial to good causes		

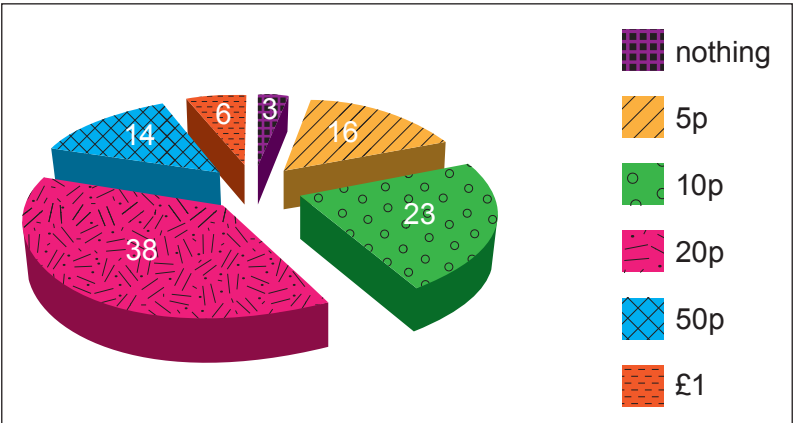


- (ii) How many fewer plastic bags will each person in the UK use over a 5-year period, based on the estimated number of bags used per person before and after the charge was introduced? [2]

Number of bags =

- (b) More than half of all consumers still regularly buy plastic bags.

The pie chart shows the results of a survey where 100 consumers were asked what is the most they would pay for a plastic bag.



Give the number of consumers that would be prepared to pay **more than** 5p for a plastic bag. [2]

Number of consumers =

- (c) Give **one** property of plastics which leads to long-term environmental problems. [1]

.....

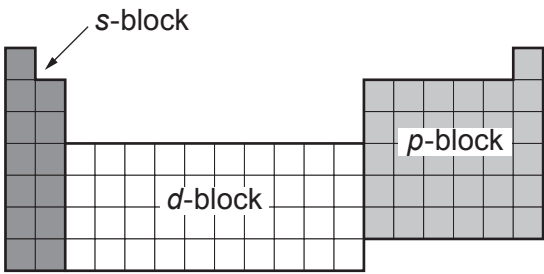
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4. The Transition Metals

The elements in the Periodic Table can be divided into three blocks: the *s*-block, the *d*-block and the *p*-block.

The *s*-block and *d*-block contain only metallic elements. The *p*-block contains both metallic and non-metallic elements.



The transition metals are found within the *d*-block and are the metallic elements that serve as a bridge or ‘transition’ between the two sides of the table. They include the elements iron, silver, gold, vanadium and titanium.

The transition metals have similar properties to each other. These include high melting point, high density, good conductivity and malleability. The transition metals also have unique properties, making them different from the main group metals, including the ability to form compounds with different colours and different oxidation states.

Table 1 gives the properties of some transition metals and some Group 1 metals.

Element	Melting point (°C)	Boiling point (°C)	Oxidation states	Colours of compounds formed
iron	1536	2861	+2, +3	green and brown
lithium	180	1342	+1	white
potassium	63	760	+1	white
sodium	98	883	+1	white
titanium	1660	3287	+3, +4	violet and white
vanadium	1910	3407	+3, +4, +5	green, blue and yellow

Table 1



- (a) Tick (✓) the statement that **best** describes where metals are positioned within the Periodic Table. [1]

the s-block, *p*-block and *d*-block

☐

the s-block and *p*-block only

☐

the s-block only

☐

the *d*-block and *p*-block only

☐

the *p*-block only

☐

- (b) Name the element in **Table 1** that is a liquid over the biggest temperature range. [1]

.....

- (c) Tick (✓) the **two** statements that correctly describe the oxidation states of the metals listed in **Table 1**. [2]

the Group 1 metals and transition metals all have a +1 oxidation state

☐

the transition metals all have a +3 oxidation state

☐

the Group 1 metals all have a +1 oxidation state

☐

iron and lithium have the same oxidation states

☐

the Group 1 metals and transition metals all have a +4 oxidation state

☐

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(d) Consider the following statement.

‘It is possible to tell the Group 1 and transition metals apart by the colour of the compounds they form.’

Use the information in **Table 1** to give **one** reason to agree and **one** reason to disagree with this statement. [2]

Reason to agree

.....

.....

Reason to disagree

.....

.....

6

